Degrees of Uncomputability: Post's problem and the Finite Injury Priority method

Master's Degree in Computer Science

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Roadmap

- Introduction on computation
- Degrees of Uncomputability
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What is computation?

Defining computation

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- *Successor function*: on input x return $x + 1$
- *Projection function*: on input $(x_1, \ldots, x_n)$ return x_i
- *Composition operator*: on input $h, g_1, \ldots, g_m$ return the function that applies h together on the outputs of the applications of $g_1, \ldots, g_m$
- *Primitive recursion operator*: on input h, g, return the function that applies g if the first input is 0 and otherwise applies h on the first argument x_0 and then recursively applies the process on the predecessor of x.
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- *Constant functions*: $C_n^k(x_1, \ldots, x_k) := n$ for all $k, n \in \mathbb{N}$
- *Successor function*: $S(x) := x + 1$
- *Projection function*: $P_i^k(x_1, \ldots, x_k) := x_i$ for all $k \in \mathbb{N}$
- *Composition operator*:
  $(h \circ (g_1, \ldots, g_m))(x_1, \ldots, x_k) := h(g_1(x_1, \ldots, x_k), \ldots, g_m(x_1, \ldots, x_k))$
- *Primitive recursion operator*: $\rho(g, h) := f$ where:

$$f(x_0, x_1, \ldots, x_k) = \begin{cases} g(x_1, \ldots, x_k) & \text{if } x_0 = 0 \\ h(y_0, f(y_0, x_1 \ldots, x_k)) & \text{if } x_0 = S(y_0) \end{cases}$$

- *Minimization operator*: $\mu(f) := f'$ where:

$$f'(x_1, \ldots, x_k) = \begin{cases} z & \text{if } f(z, x_1, \ldots, x_k) > 0 \text{ and } f(i, x_1, \ldots, x_k) > 0 \text{ for all } i < z \\ * & \text{otherwise} \end{cases}$$

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1935 – Alonzo Church defines λ**-calculus** to settle the Entscheidungsproblem problem.

- *Language grammar*: $M, N ::= x \mid \lambda x.M \mid MN$
- *Substitution*: $M[x := N]$ means that x gets substituted by N in M
- α*-rule*: $\lambda x.M \xrightarrow{\alpha} \lambda y.(M[x := y])$
- β*-rule*: $(\lambda x.M)N \xrightarrow{\beta} M[x := N]$
- η*-rule*: $\lambda x.Mx \xrightarrow{\eta} M$ if $x \notin \text{free}(M)$

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Defining computation

1936 – Alan Turing defines a simple abstract model of a computer.

- Infinite tape subdivided in cells with a read-write head
- Set of instructions defined by a function δ
- Each instruction moves the head left or right based on the current state and the value read.
- It may also change the value below the head before moving.

Turing's machine also settles the Entscheidungsproblem and it is equivalent to λ-calculus. Recognized by Gödel himself as the best definition of computation.

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$\Phi_{i,s}(x)$ denotes the **computation** of the i-th TM for $s \in \mathbb{N}$ steps on input $x \in \mathbb{N}$.

- $\Phi_{i,s}(x) \downarrow$: the computation halts after s steps
- $\Phi_i(x) \downarrow$: there is a $s \in \mathbb{N}$ such that $\Phi_{i,s}(x) \downarrow$
- $\Phi_i(x) \uparrow$: there is no $s \in \mathbb{N}$ such that $\Phi_{i,s}(x) \downarrow$
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Computability

Definitions

Given a set $S \subseteq \mathbb{N}$, we say that S is:

- **Semi-decidable** when $\exists i \in \mathbb{N}$ such that $\forall x \in \mathbb{N}$ it holds that $\psi_i(x) = 1$ if $x \in S$, otherwise $\psi_i(x) = 0$ or $\Phi_i(x) \uparrow$.
- **Decidable** when $\exists i \in \mathbb{N}$ such that $\forall x \in \mathbb{N}$ it holds that $\psi_i(x) = 1$ if $x \in S$, otherwise $\psi_i(x) = 0$.
- **Recursively enumerable** when there is an algorithmic procedure $A : \mathbb{N} \to \{0, 1\}$ such that $S = \{A(0), A(1), A(2), \ldots\}$

Obs. 1: S is semi-decidable if and only if it is recursively enumerable
Obs. 2: S is decidable if and only if both S and $\overline{S}$ are semi-decidable.

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Introduction

Turing's work

Turing [Tur36] proved that:

- Some sets that are semi-decidable but undecidable (e.g. $H = \{\langle i, x\rangle \mid \Phi_i(x) \downarrow\}$).
- Some sets cannot be semi-decided (e.g. $\overline{H}$).

Three degrees of computability: **solvable** problems, **semi-solvable** problems and **unsolvable** problems.

Are there some other degrees of computability?

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2 Degrees of uncomputability

Degrees of uncomputability

Turing degrees

Post [Pos44] formalized the idea of computability degrees through **Turing reductions**.

Let Φ_i^B denote the Turing machine Φ_i with access to an oracle machine that decides B.

- *Turing reducibility*: $A \leq_T B$ when $\exists i \in \mathbb{N}$ such that $\Phi_i^B(x) \downarrow$ for all $x \in \mathbb{N}$ and $A = \{x \in \mathbb{N} \mid \psi_i^B(x) = 1\}$.
- *Turing equivalence*: $A \equiv_T B$ when $A \leq_T B$ and $B \leq_T A$
- The set $\mathcal{D} = 2^{\mathbb{N}}/_{\equiv_T}$ is the set of **Turing degrees**.
- $[A]$ is the class containing A over $\mathcal{D}$.

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We say that $[A]$ is **lower** than $[B]$, written as $[A] \preceq [B]$, if $A \leq_T B$.

$\preceq$ forms an ordering over the set $\mathcal{D}$, giving an hierarchy of unsolvability degrees.

$0 = [\varnothing]$ is the unique degree containing all decidable problems

- There is no degree below 0
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$0 = [\varnothing]$ is the unique degree containing all decidable problems

- There is no degree below 0
- 0 is lower than any degree

Degrees of uncomputability

The jump operator

The strict relation $[A] \prec [B]$ between degrees can be easily forced through **Turing jumps**.

- Given a set $X \subseteq \mathbb{N}$, the *Turing jump* of X is the set $X' = \{\langle i, x\rangle \mid \Phi_i^X(x) \downarrow\}$
  - $X \leq_T X'$ but $X' \not\leq_T X$ since X' is a variant of the Halting problem using X as an oracle.
- The jump of $\varnothing$ is exactly the Halting problem H
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Low degrees

- The jump operator can be iteratively applied to get infinite levels of the hierarchy
- All r.e. degrees are below $0'$
  - A degree is said to be r.e. if it contains an r.e. set
  - Not all sets inside an r.e. degree are r.e. (e.g. $\overline{H} \in 0'$)
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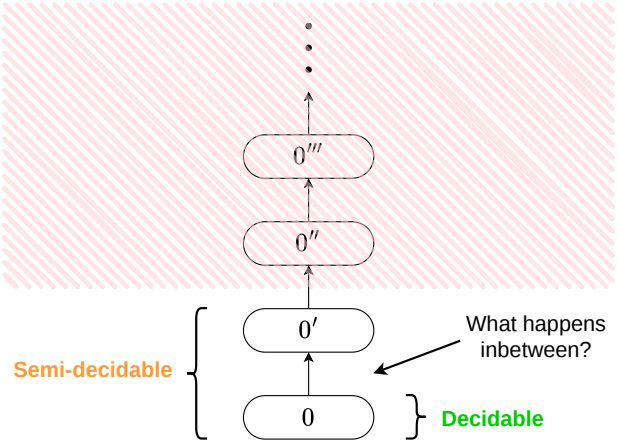

Degrees of uncomputability

Post's problem

- In [Pos44], Emil Post investigates the existence of a degree d such that $0 \prec d \prec 0'$
- He was able to prove that the existence of such degree,
- However, he couldn't find a way to force the degree to be r.e.

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- Post's problem was solved 12 years later independently by Friedberg [Fri57] and Muchnik [Muc56] through the **finite injury priority method**.
- The method is an improvement on the **finite extension method** developed by Post in his original works
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The Finite Extension method

The idea

- We define a list of requirements $\{R_i\}_{i\in\mathbb{N}}$ that will enforce our desired property over the sets A and B
- We construct two list of strings $\{\alpha_i\}_{i\in\mathbb{N}}$ and $\{\beta_i\}_{i\in\mathbb{N}}$ where:
  - α_i is a finite extension of α_{i-1} (same goes for β_i and β_{i-1})
  - The strings α_i, β_i will satisfy requirement R_i
  - Strings are to be considered as partial functions from $\mathbb{N}$ to $\{0, 1\}$ (equivalently, as an infinite tape over $\{0, 1, *\}$)
- The sets $A = \bigcup_{i=0}^{+\infty} \mathbb{N}[\alpha_i]$ and $B = \bigcup_{i=0}^{+\infty} \mathbb{N}[\beta_i]$ will have our desired property
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The Kleene-Post theorem

Theorem

[KP54] There are two low sets A and B that are incomparable

- Requirements:

$$R_{2j} := (\psi_j^B \neq A) \qquad R_{2j+1} := (\psi_j^A \neq B)$$

- Start by setting $\alpha_0 = \beta_0 = \varepsilon$

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- If there is any finite extension α' of α_{2j-1} such that $\Phi_j^{\{\alpha'\}}(x) \downarrow$, we set:

$$\alpha_{2j} = \alpha' \qquad \beta_{2j}(y) = \begin{cases} 1 - \psi_j^{\{\alpha'\}}(x) & \text{if } y = x \\ \beta_{2j-1}(y) & \text{otherwise} \end{cases}$$

- Otherwise, we set:

$$\alpha_{2j} = \alpha_{2j-1} \qquad \beta_{2j}(y) = \begin{cases} \text{rand}(\{0,1\}) & \text{if } y = x \\ \beta_{2j-1}(y) & \text{otherwise} \end{cases}$$

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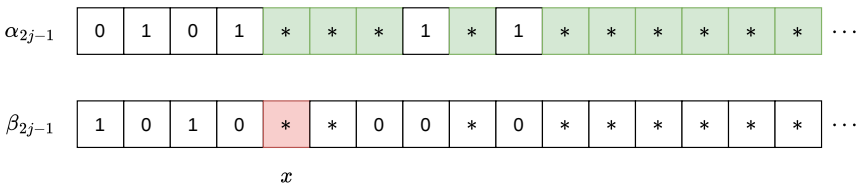
Is there a way to set a finite number of these bits in a way such that $\Phi_{2j}^{\alpha'}(x)$ terminates?



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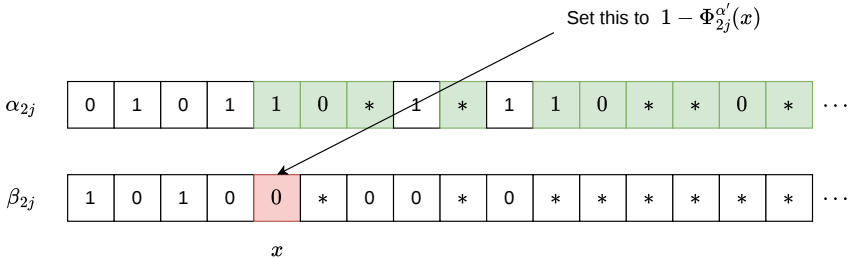

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- Strings α_{2j}, β_{2j} satisfy requirement R_{2j}
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The Finite Injury Priority method

The idea

- **Injury**: a requirement gets injured when its satisfaction is not guaranteed anymore.
- The key idea is to assign a priority level to each requirement:
  1. Each requirement will have finitely many requirements with higher priority
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The Finite Injury Priority method

The query function

- To keep track of injuries, we'll use a **query function** $\omega^A_{i,s}(x)$ that returns the largest index queried by $\Phi^A_{i,s}(x)$.

$$\omega^A_{i,s}(x) = \begin{cases} \max\{z \in \mathbb{N} \mid z \text{ gets queried in } \Phi^A_{i,s}(x)\} & \text{if } \Phi^A_{i,s}(x) \downarrow \\ -1 & \text{otherwise} \end{cases}$$

where $A \subseteq \mathbb{N}$ and $i, s, x \in \mathbb{N}$.

The Finite Injury Priority method

The Friedberg-Muchnik theorem

Theorem

[Fri57; Muc56] There are two r.e. sets A and B that are incomparable

- Requirements:

$$R_{2j} := (\psi_j^B \neq A) \qquad R_{2j+1} := (\psi_j^A \neq B)$$

- We say that an index x is a **witness** for the requirement R_{2i} at step s if $\Phi_{i,s}^{B_s}(x) \downarrow$ and $\psi_{i,s}^{B_s}(x) \neq A_s(x)$ (symmetric definition for R_{2i+1})

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- On each step $s \in \mathbb{N}$, for each $j \in \mathbb{N}$ we compute:
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- **Invariants**:
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- Let $A_0 = B_0 = \varnothing$ and let $w_{j,0} = r_{j,0} = -1$ for each $j \in \mathbb{N}$.
- Consider a generic step $s > 0$
- For each $j \leq s$, if $w_{j,s-1} \neq -1$ we propagate the previous values because R_{2i} already has a witness.
  - $A_s = A_{s-1}$ and $B_s = B_{s-1}$
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The Finite Injury Priority method

The Friedberg-Muchnik theorem

- Let $A_0 = B_0 = \varnothing$ and let $w_{j,0} = r_{j,0} = -1$ for each $j \in \mathbb{N}$.
- Consider a generic step $s > 0$
- For each $j \leq s$, if $w_{j,s-1} \neq -1$ we propagate the previous values because R_{2i} already has a witness.
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- Suppose now that $w_{2h,s-1} = -1$.
- Let x be the minimum index such that $x \notin A_{s-1}$ and such that for all $k < 2h$ it holds that $x > r_{k,s}$.
- If $\Phi_{h,s-1}^{B_{s-1}}(x) \uparrow$, we preserve that $w_{2h,s} = r_{2h,s} = -1$ and propagate $A_s = A_{s-1}$, $B_s = B_{s-1}$. This trivially satisfies the requirement R_{2h}.
- Otherwise, if $\psi_{h,s-1}^{B_{s-1}}(x) = 0$, we set:

$$w_{2h,s} = x \qquad r_{2h,s} = \max(x, \omega_{h,s-1}^{B_{s-1}}(x))$$

$$A_s = A_{s-1} \cup \{x\} \qquad B_s = B_{s-1}$$

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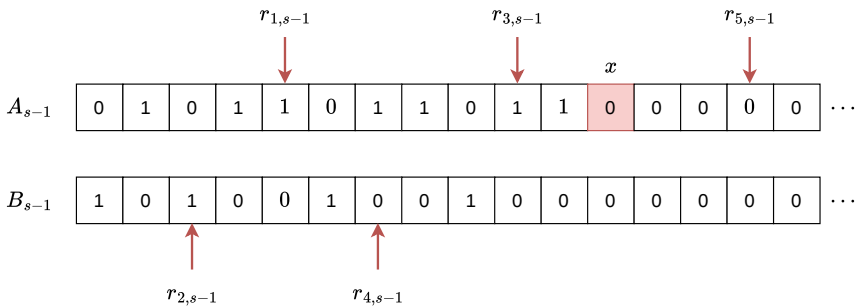
What happens when $w_{4,s} = -1$?



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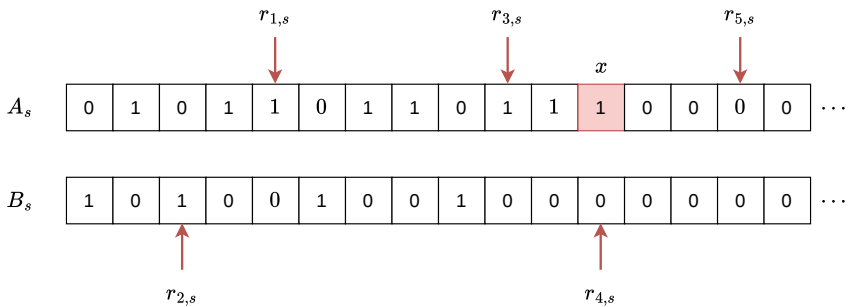

The Finite Injury Priority method

The Friedberg-Muchnik theorem

Claim: For each $j \in \mathbb{N}$ it holds that:

1. R_j is injured at most $2^j - 1$ times
2. There is a step s_0 such that for all $s \geq s_0$ and for all $k < j$ no new index is added to A_s

The Finite Injury Priority method

The Friedberg-Muchnik theorem

- $A = \bigcup_{s=0}^{+\infty} A_s$ and $B = \bigcup_{s=0}^{+\infty} B_s$
- The claim implies that every requirement will be, eventually, satisfied by the construction.
- Moreover, the use of the restriction values $r_{j,s}$ allows us to recursively enumerate the sets A and B by restricting our interest to the indexes between 0 and $r_{j,s}$ for each $j \in \mathbb{N}$
  - This implies that A and B are both r.e.!

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The Finite Injury Priority method

Other interesting results

- **Infinite Injury Priority Argument**: Sacks [Sac64] proved that the above construction can be extended to a countably infinite argument
- **Density of r.e. sets**: For each pair of r.e. sets A, B there is another r.e. set C such that $A <_T C <_T B$
- Simpson [Sim77] proved that the first-order theory of $\mathcal{D}$ over the language $(\preceq, =)$ is many-one equivalent to the theory of true second-order arithmetic.

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Thank you for listening!
Any questions?

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